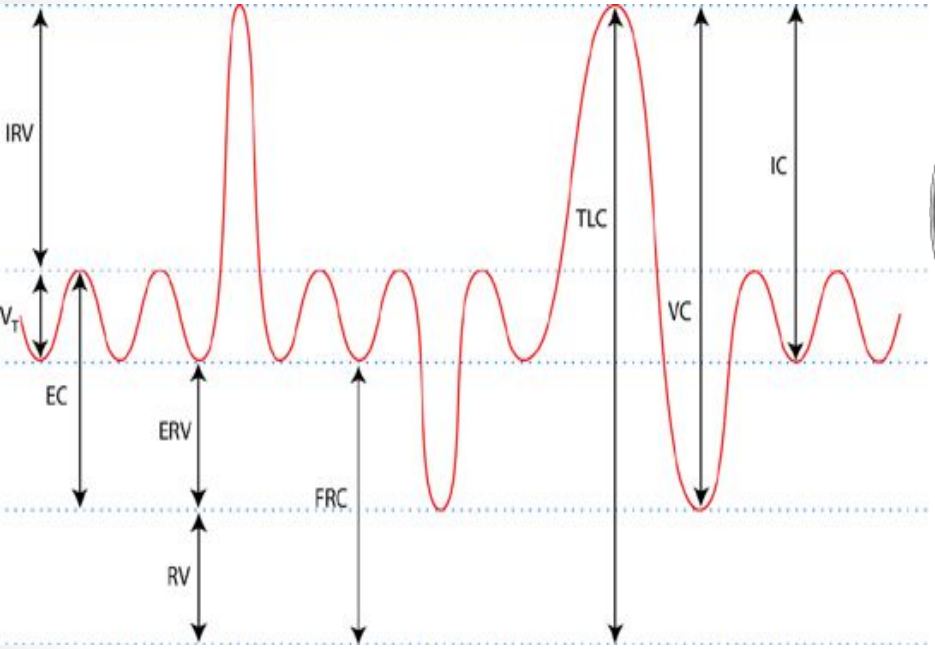
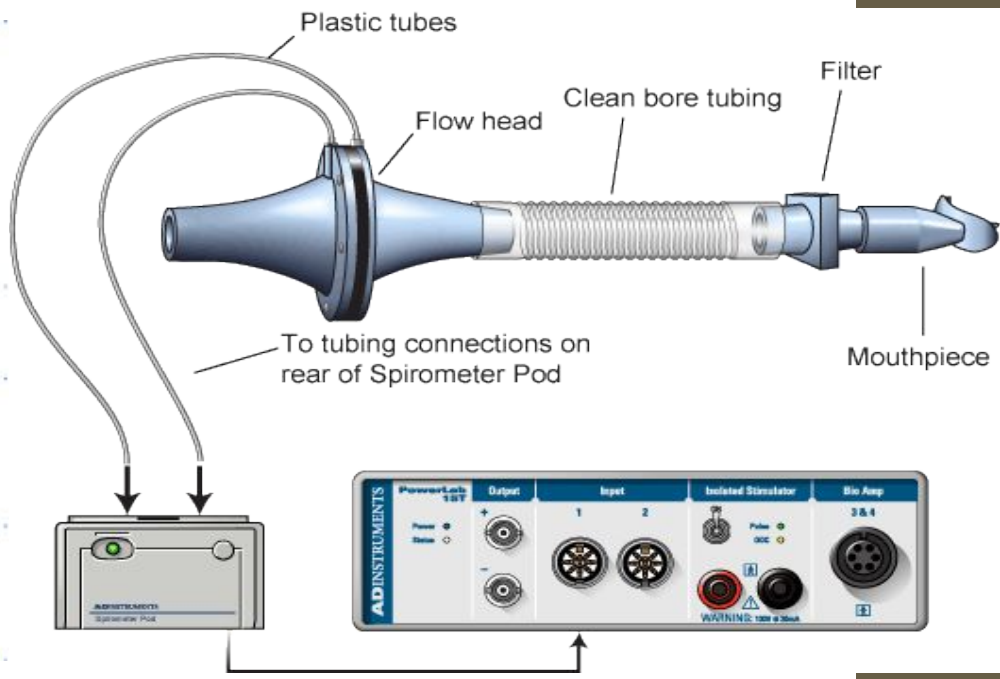
To study pulmonary function test by measuring lung volumes and capacities using Spirometer / Vitalograph / Power Lab





DR ABDUL SAMAD KHAN
SENIOR LECTURER-DIKIOS
DOW UNIVERSITY OF HEALTH SCIENCES

LEARNING OBJECTIVES

At the end of session, the students should be able to describe and give the importance of;

1.Different lung volumes including

- Tidal Volume (TV)
- Inspiratory Reserve Volume (IRV)
- Expiratory Reserve Volume (ERV)
- Residual Volume (RV).

2.Different lung capacities

Total Lung Capacity (TLC).

- Vital Capacity (VC).
- Functional Residual Capacity (FRC).
- Inspiratory Capacity (IC).

VOLUMES VERSUS CAPACITIES

There are **four volume subdivisions i.e.**

- (TV),
- (IRV)
- (ERV),
- (RV), which:

- do not overlap.
- can not be further divided.
- when added together equal total lung capacity.

Lung capacities

- (TLC),
- (VC),
- (FRC) and
- (IC)

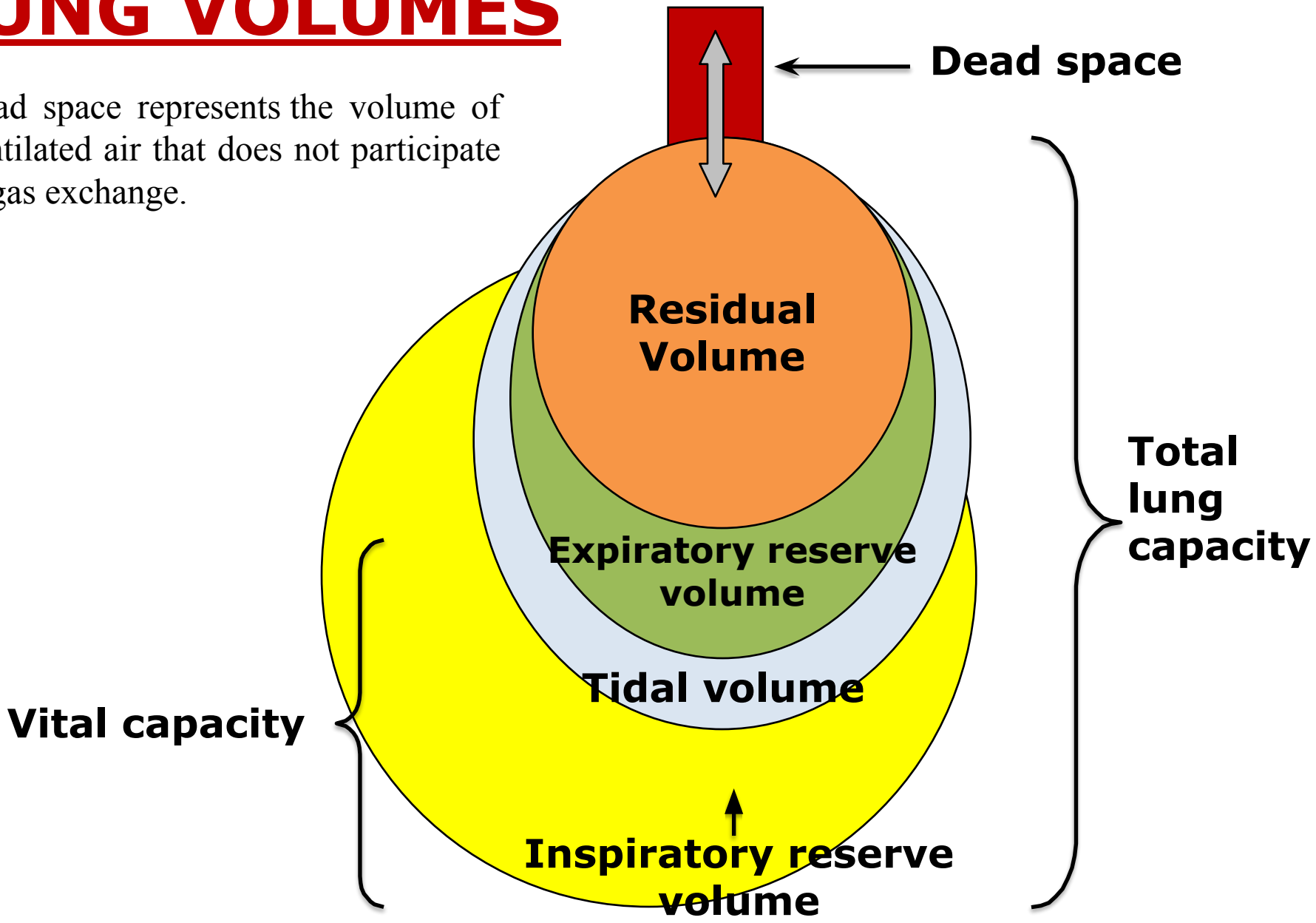
are subdivisions of total volume that include <u>two or more</u> of the 4 basic lung volumes.

BASIC LUNG VOLUMES

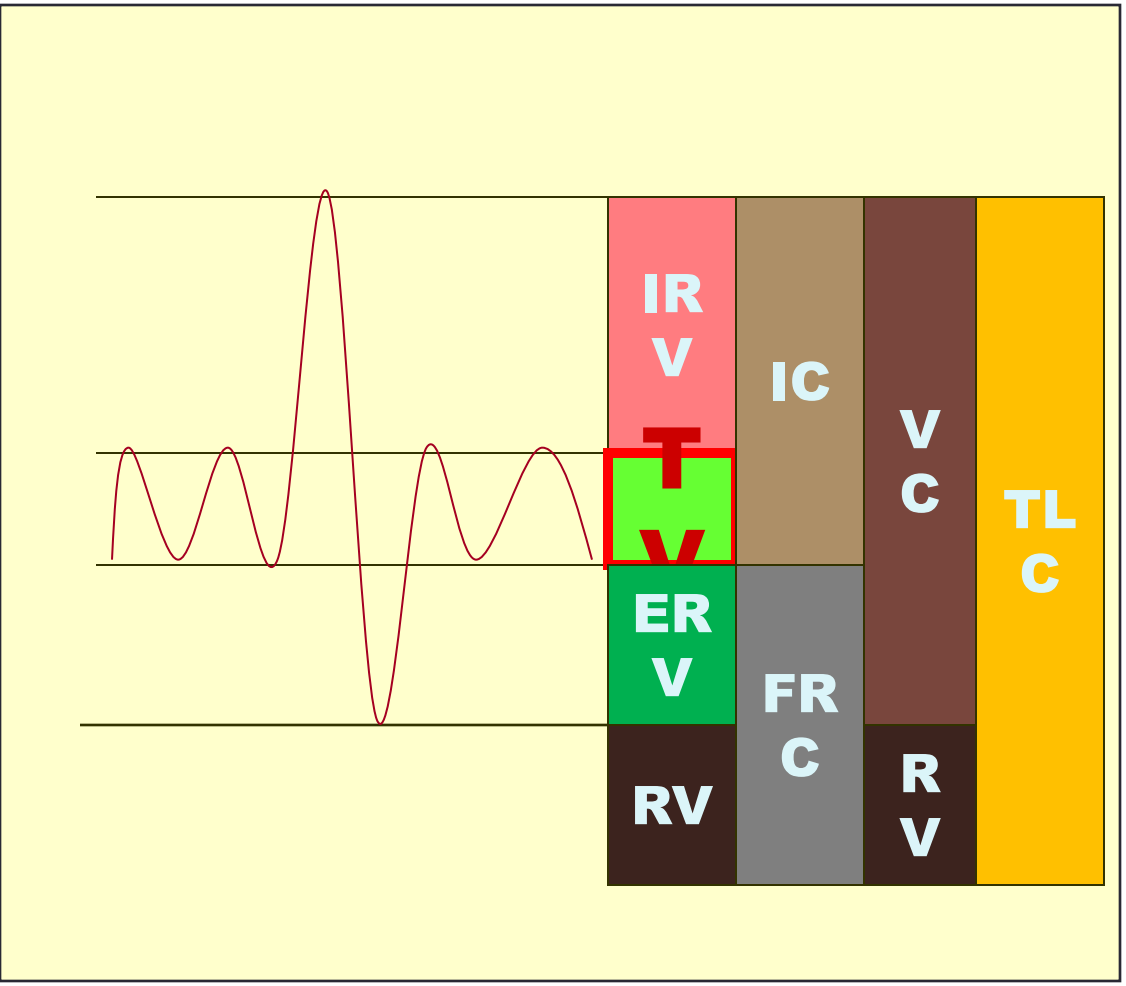
- Tidal Volume (TV)
- Inspiratory Reserve Volume (IRV)
- Expiratory Reserve Volume (ERV)
- Residual Volume (RV).

LUNG VOLUMES

Dead space represents the volume of ventilated air that does not participate in gas exchange.



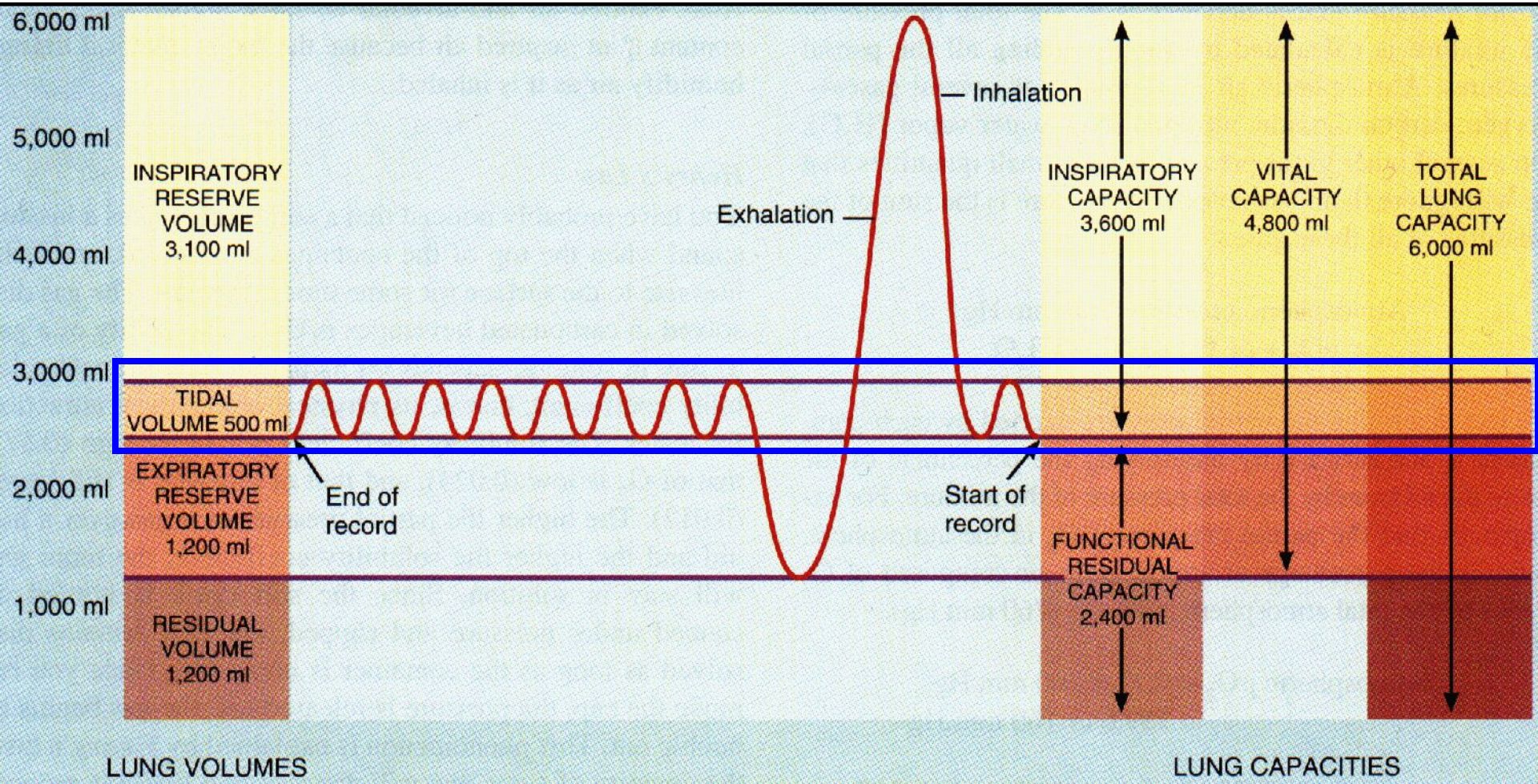
TIDAL VOLUME (TV)



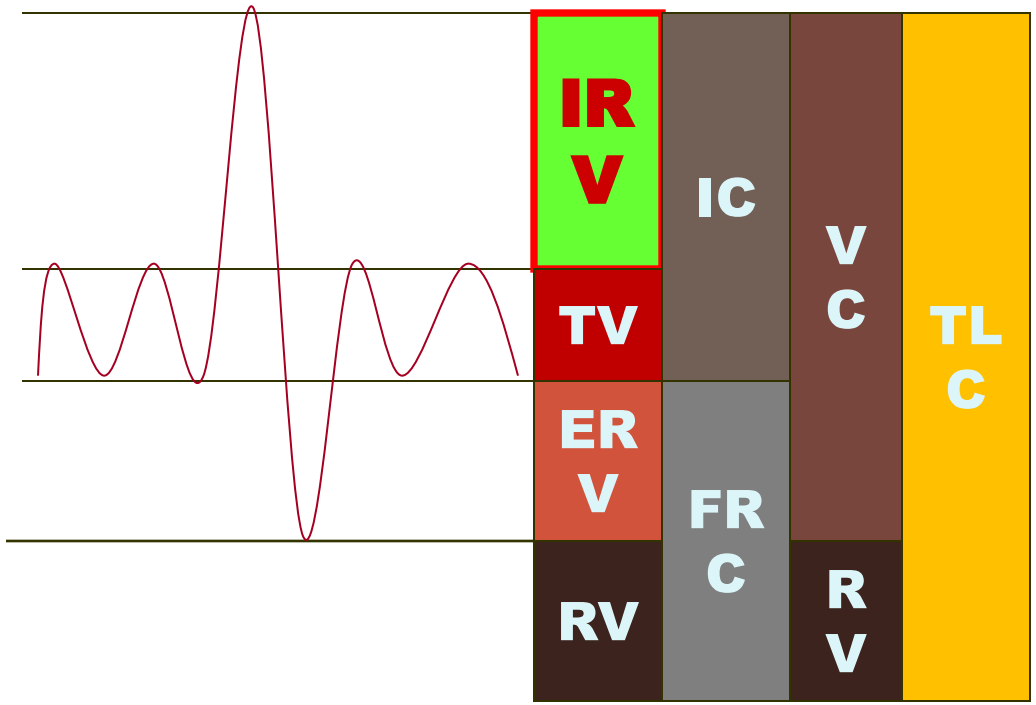
The volume of gas inspired or expired with each breath during normal quiet breathing

TIDAL VOLUME (V_T)

- volume of one breath
- about 500 ml
- only about 70% (350ml) actually reaches alveoli



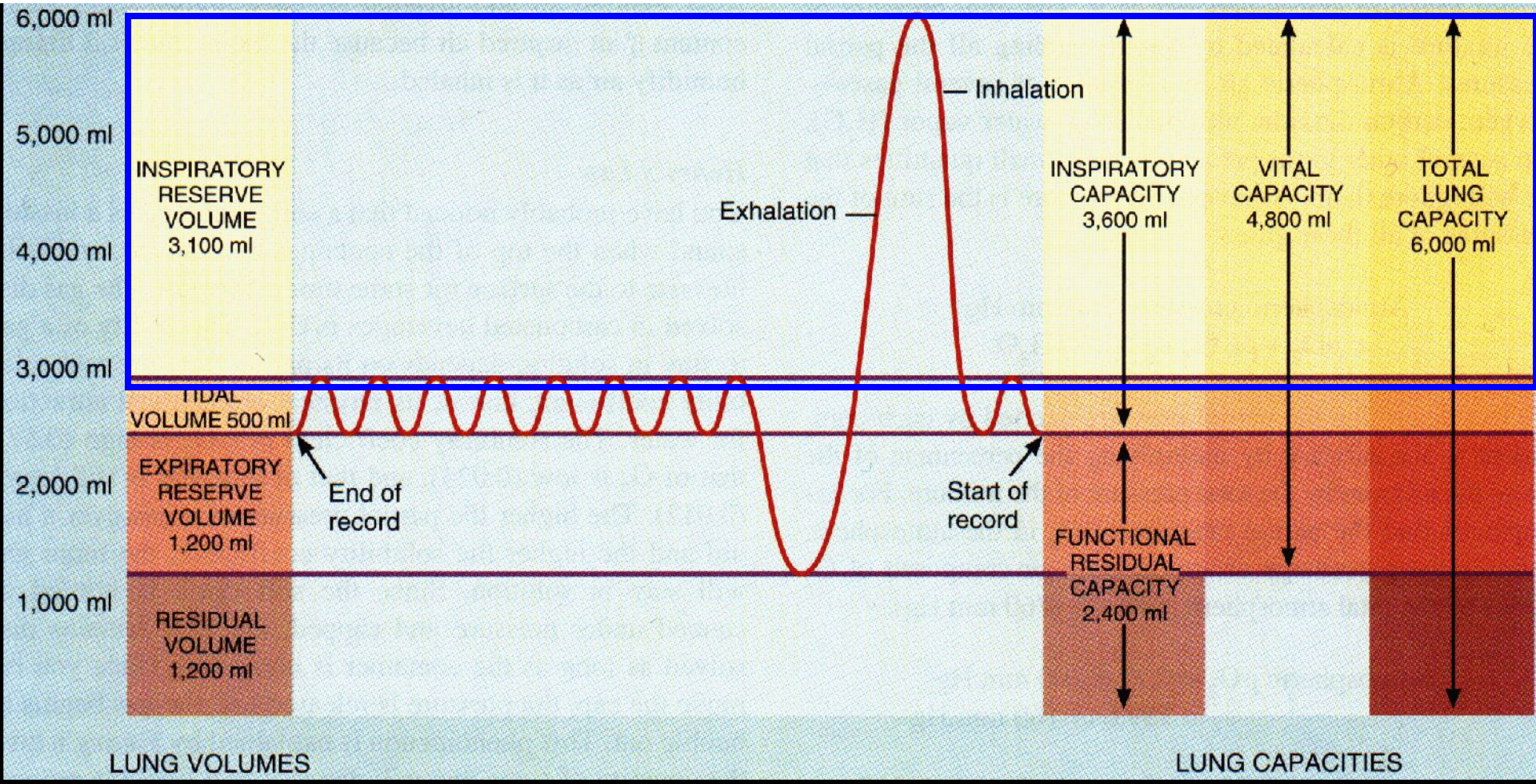
INSPIRATORY RESERVE VOLUME (IRV):



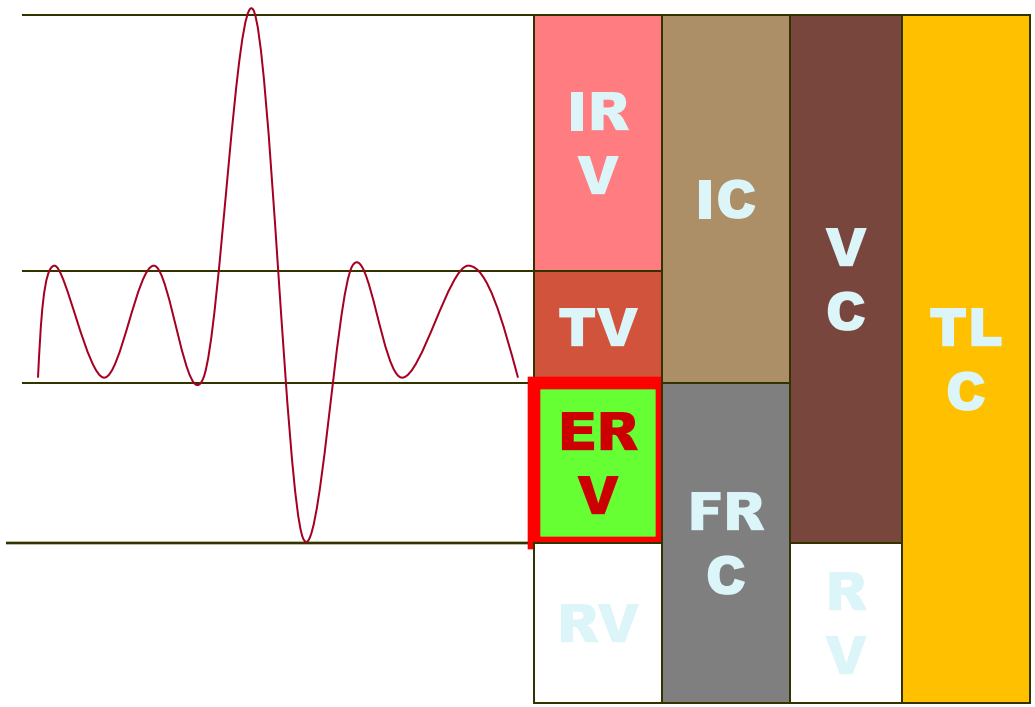
The maximum amount of air that can be inhaled after a normal tidal volume inspiration

INSPIRATORY RESERVE VOLUME (IRV):

- air inspired above normal quiet breathing
- 3100ml above 500ml V_T



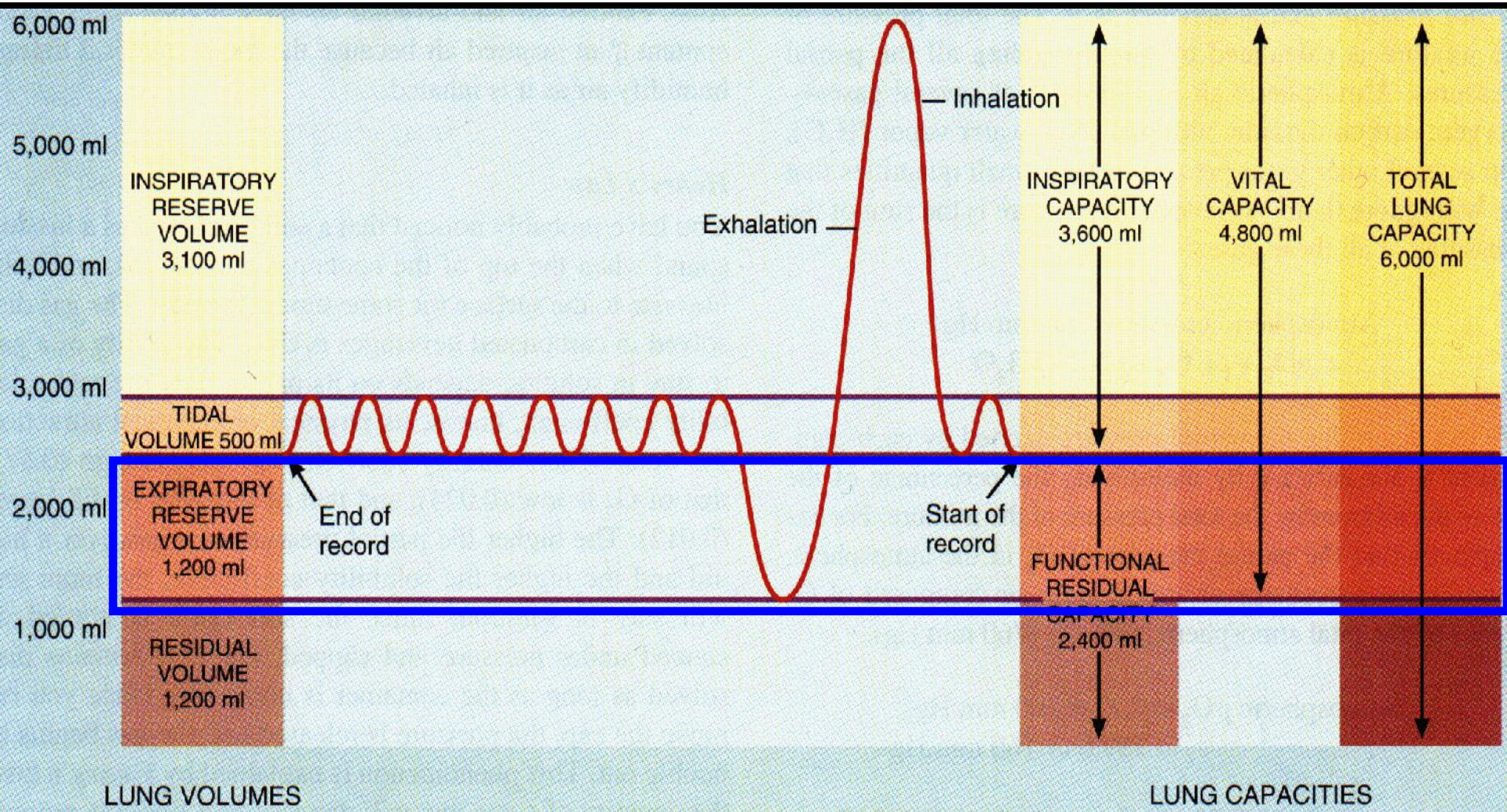
EXPIRATORY RESERVE VOLUME (ERV)



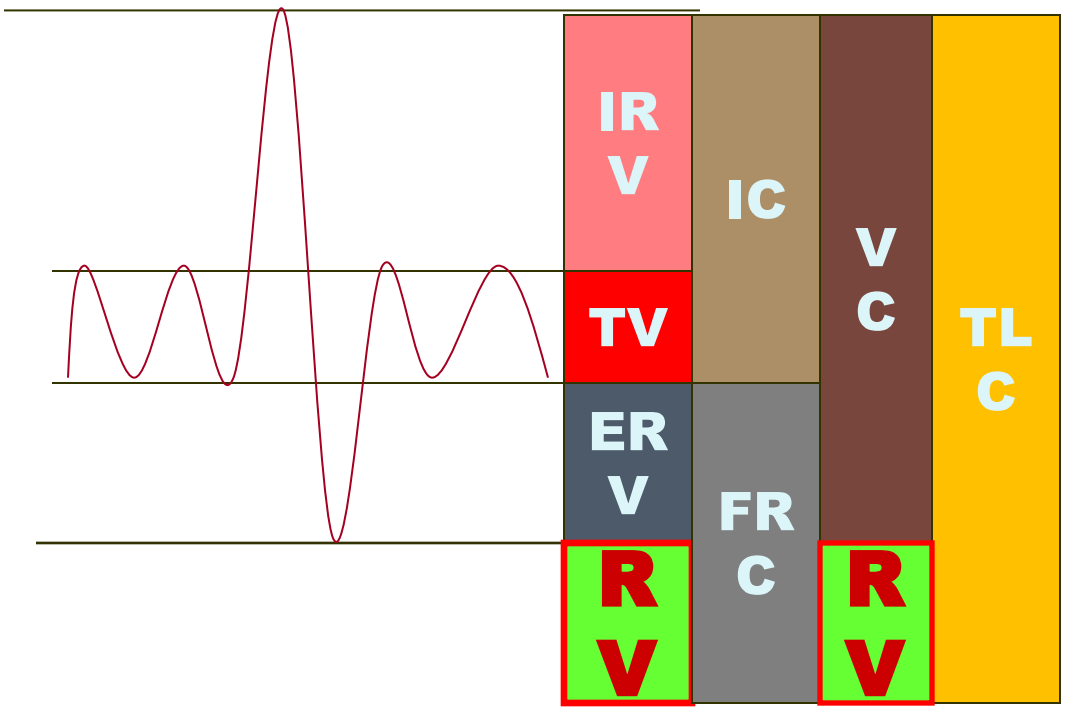
Maximum amount of air that can be exhaled from the resting expiratory level

EXPIRATORY RESERVE VOLUME (ERV)

- air expired above normal quiet breathing
- 1200 ml above 500ml V_T



RESIDUAL VOLUME (RV)



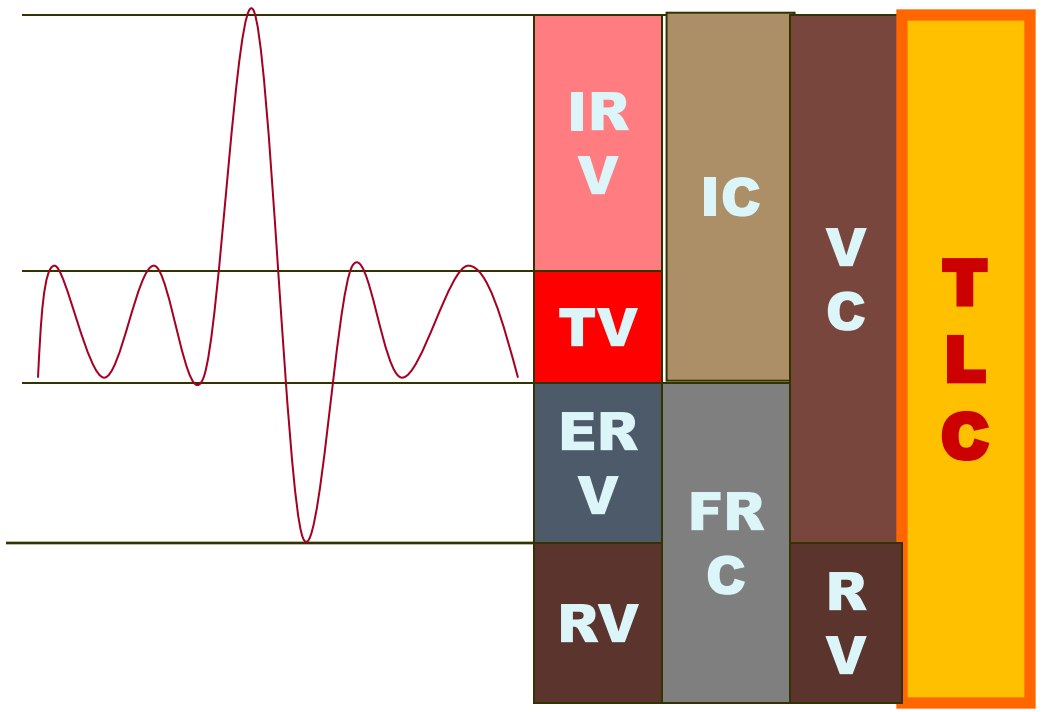
The volume of air remaining in the lung after a maximal expiration.

This is the only lung volume which cannot be measured with a spirometer.

BASIC LUNG CAPACITIES

- Total Lung Capacity (TLC).
- Vital Capacity (VC).
- Functional Residual Capacity (FRC).
- Inspiratory Capacity (IC).

TOTAL LUNG CAPACITY (TLC)

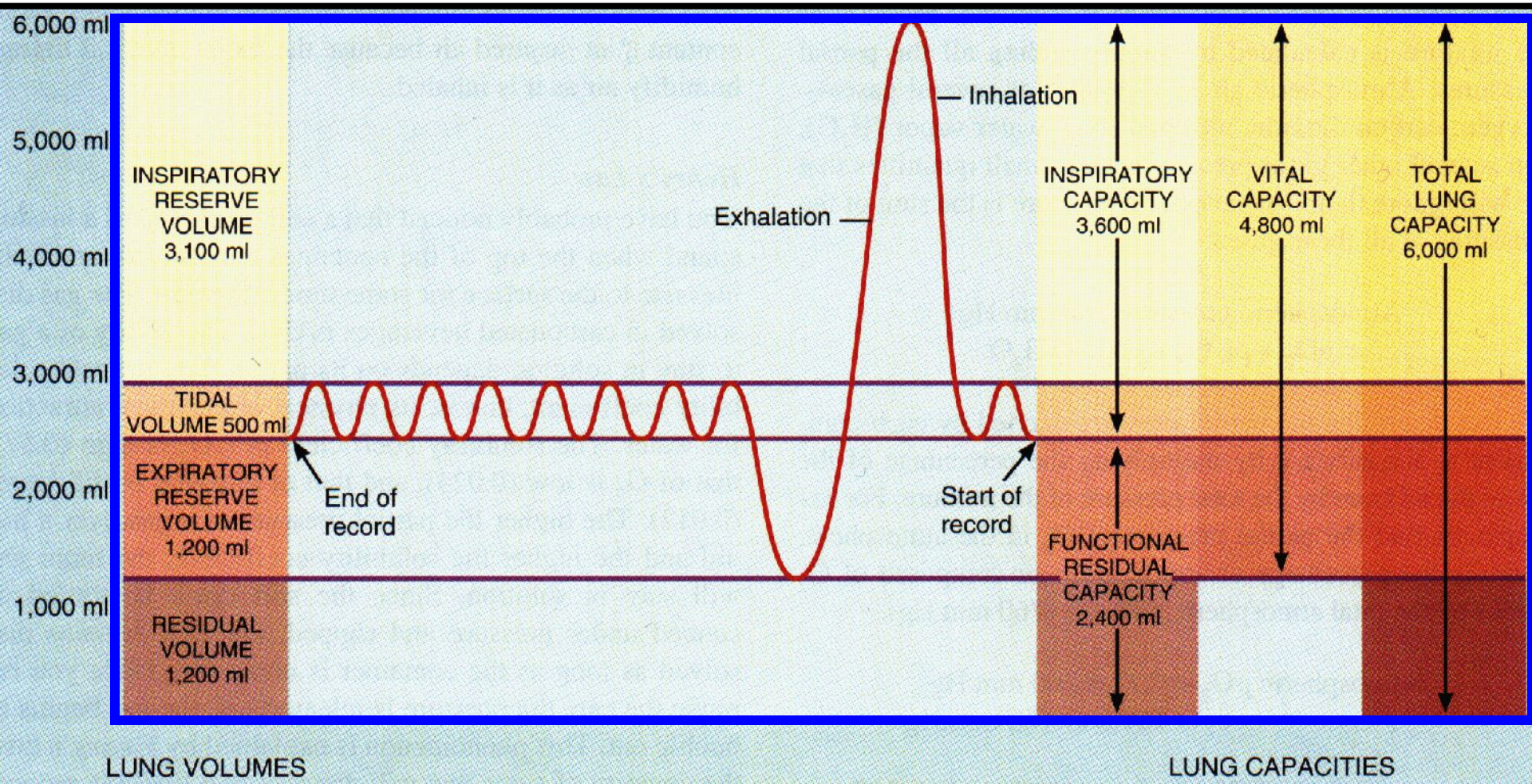


- The volume of air contained in the lungs at the end of a maximal inspiration.
- Sum of all four basic lung volumes

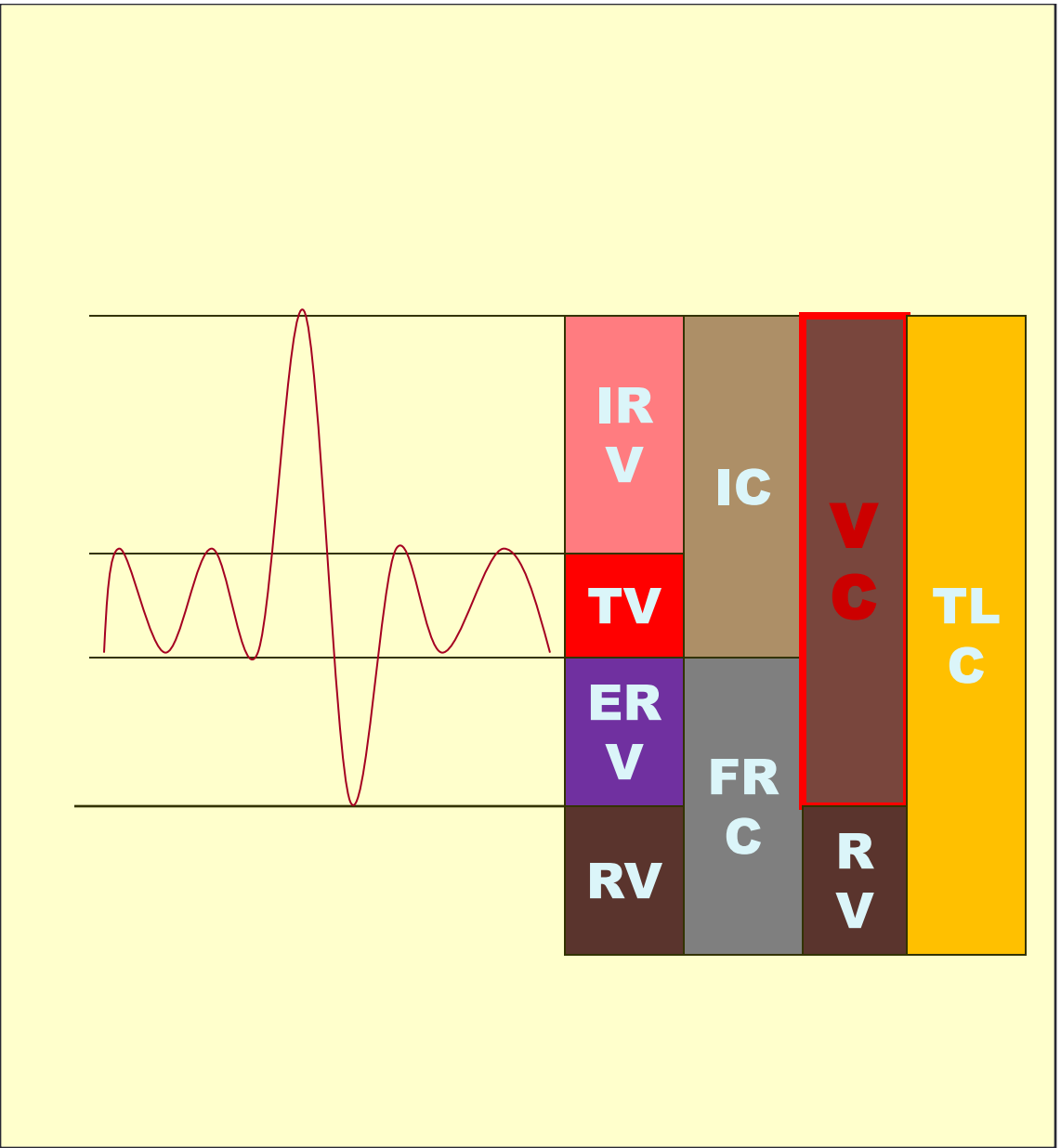
- TLC = RV + IRV + TV + ERV

TOTAL LUNG CAPACITY (TLC)

- **TLC= IRV + V_T + ERV + RV = 6000ml**



VITAL CAPACITY (VC)



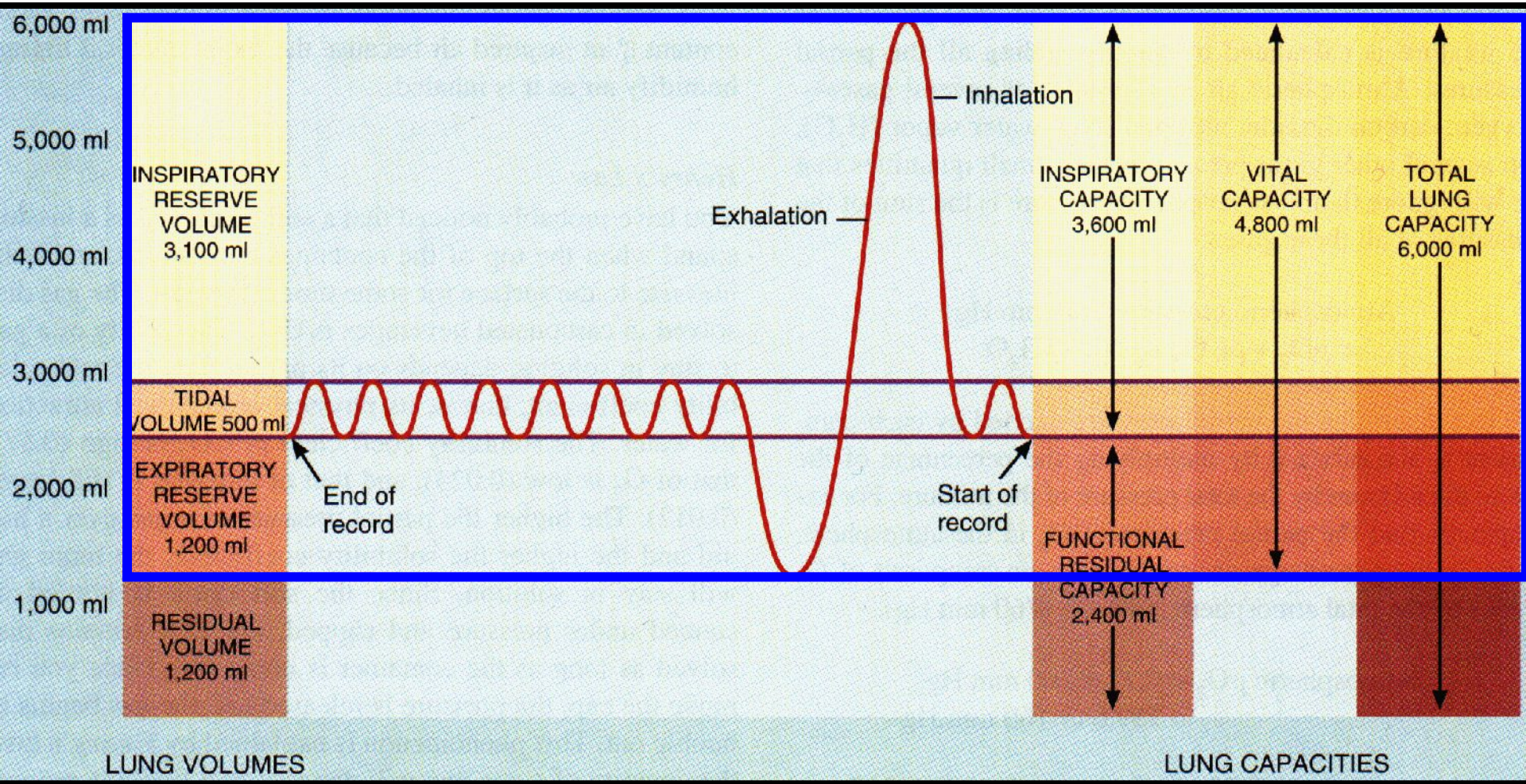
- The maximum volume of air that can be forcefully expelled from the lungs following a maximal inspiration.
- Largest volume that can be measured with a spirometer
- **VC = IRV + TV + ERV**

  OR

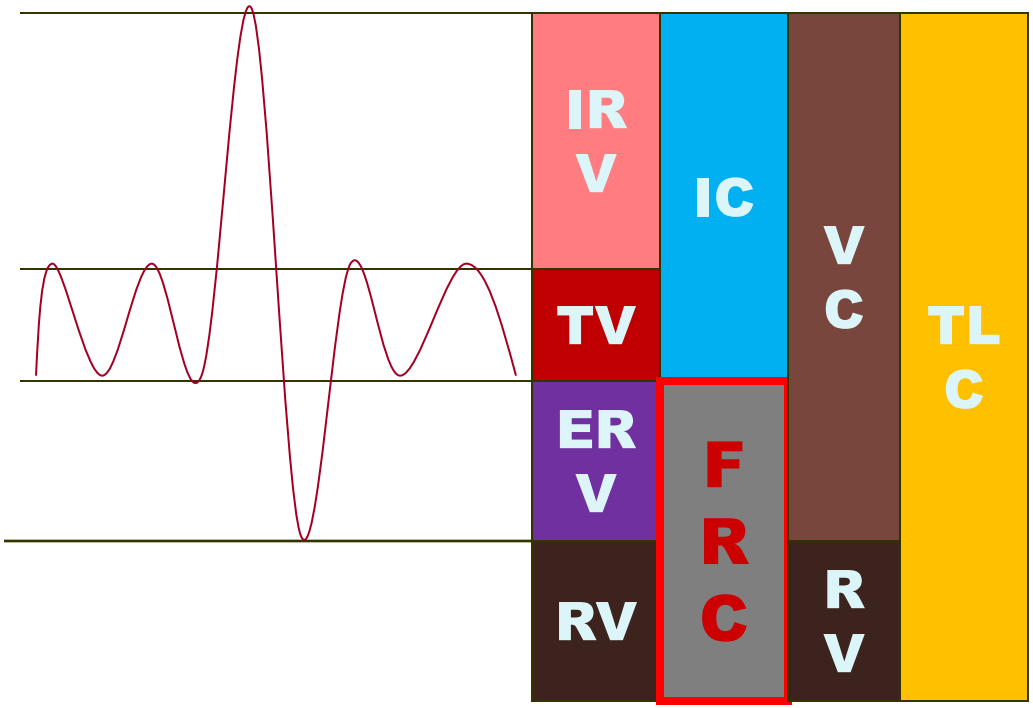
  = TLC - RV

VITAL CAPACITY (VC)

- **VC = IRV + V_T + ERV = 4800 ml**

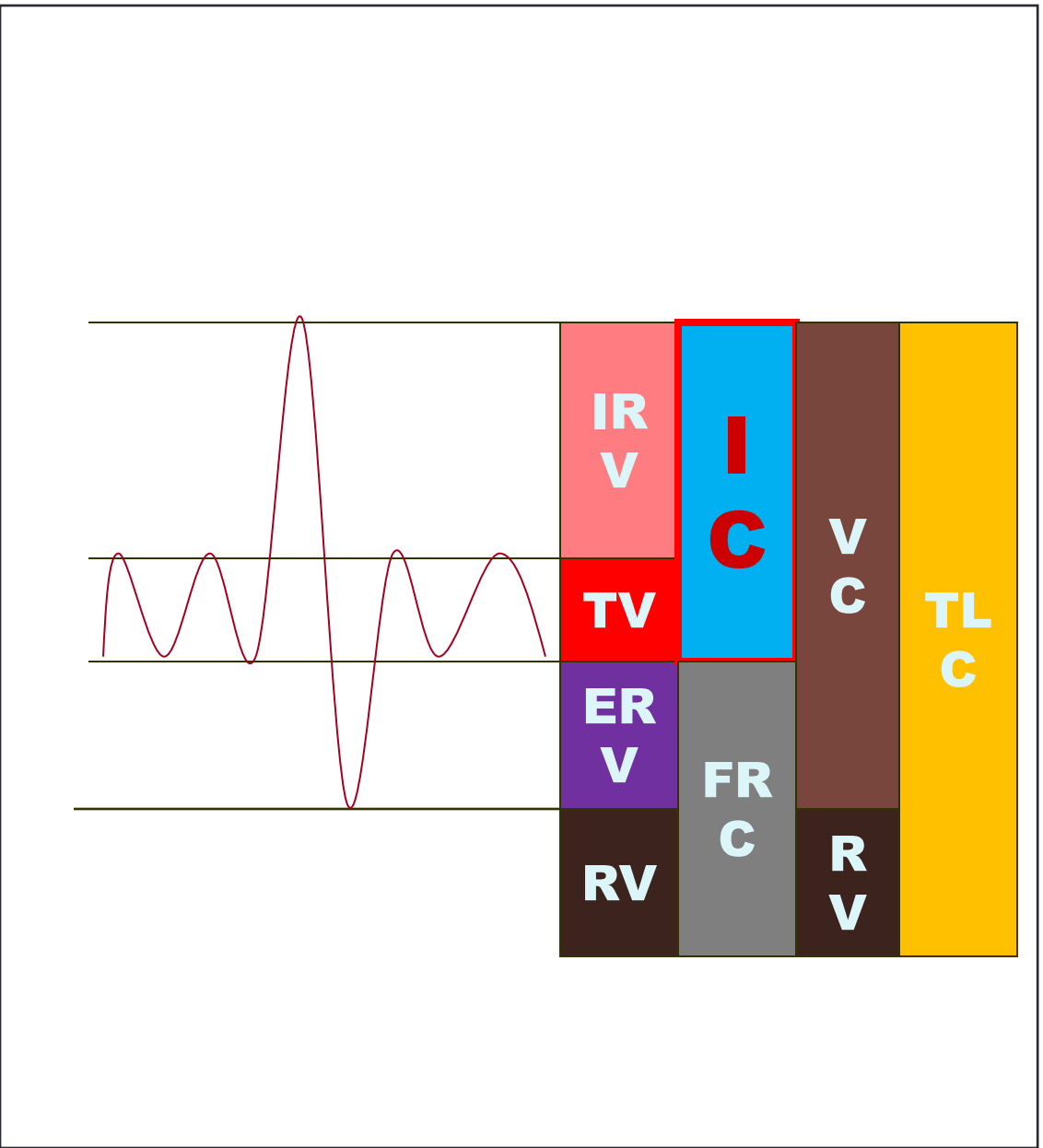


FUNCTIONAL RESIDUAL CAPACITY (FRC)



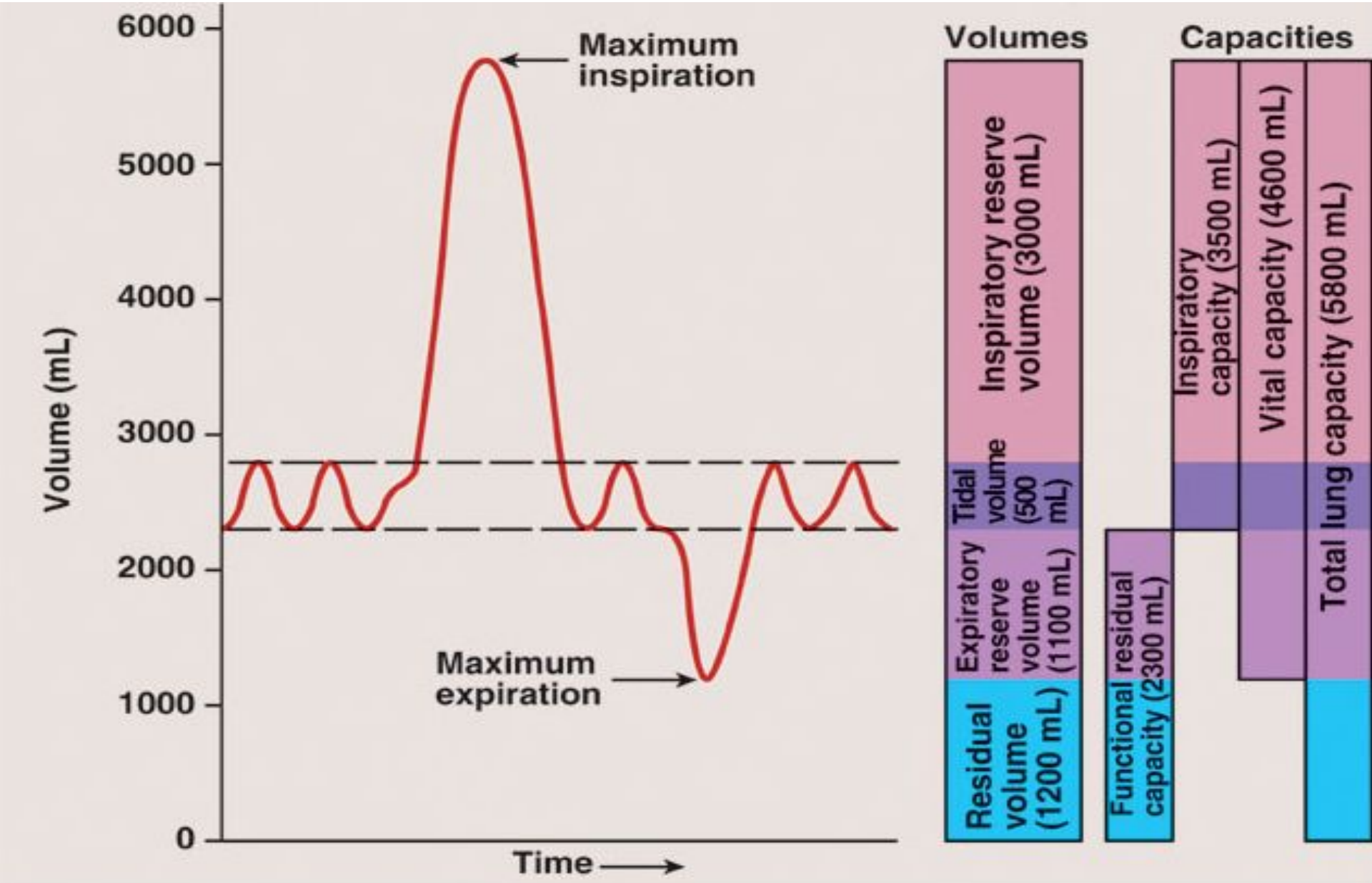
- Volume of air remaining in the lungs at the end of a normal TV expiration
- The elastic force of the chest wall is exactly balanced by the elastic force of the lungs
- **FRC = ERV + RV**

INSPIRATORY CAPACITY (IC)



- Maximum volume of air that can be inspired from end expiratory position.
- This capacity is of less clinical significance than the other three.
- **IC = TV + IRV**

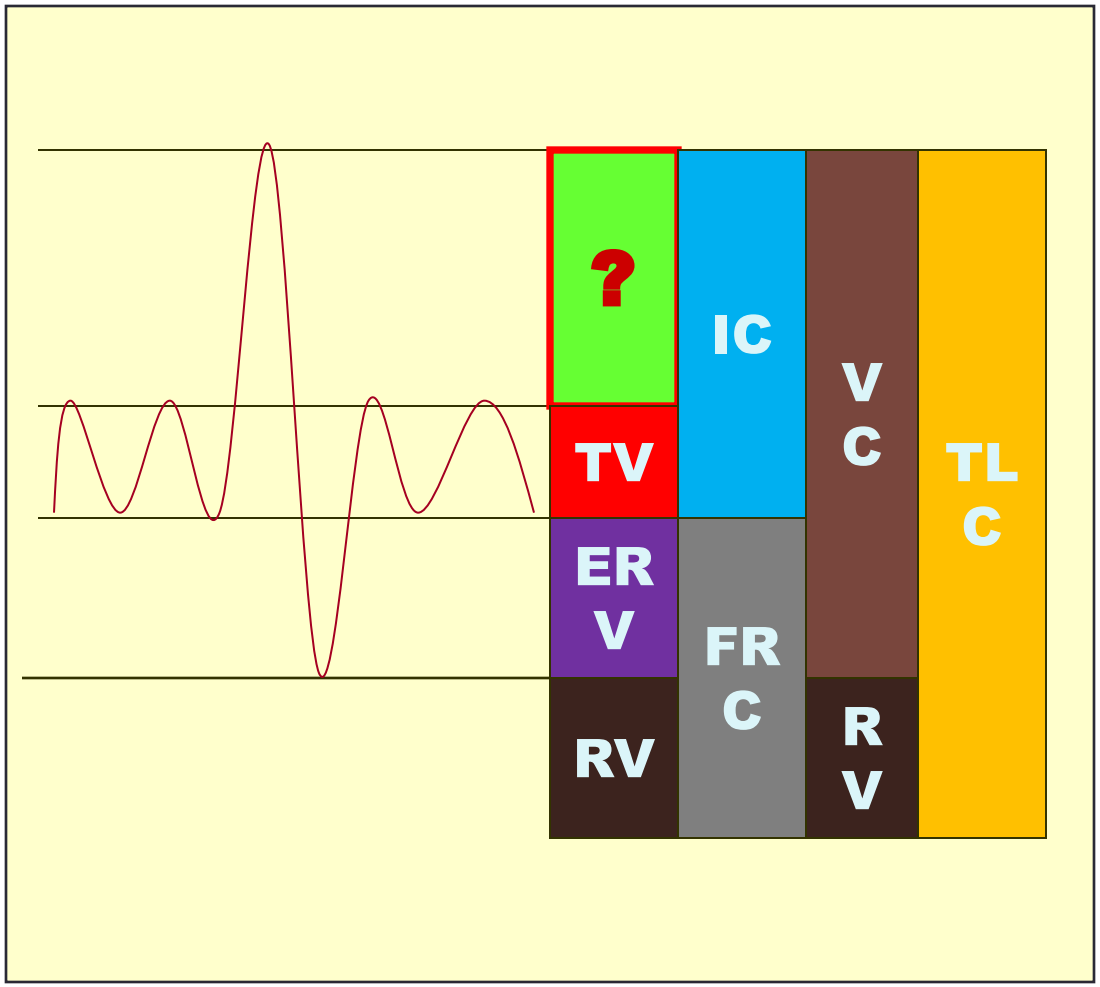
LUNG VOLUMES AND CAPACITIES



SELF ASSESSMENT

Q1. FRC = ERV + ------?

Q2. Name the volume marked in the diagram and give its value.



PRACTICAL: PULMONARY FUNCTION TESTS

OBJECT:

- To evaluate the pulmonary function test

REQUIREMENTS:

- PowerLab
- Spirometer Pod
- Respiratory flow head with connection tubes
- Clean bore tubing
- Disposable filters
- Disposable vinyl mouthpiece
- Nose clip
- Tape measure or wall chart for measuring height

PROCEDURE:

1- Re-zero the Spirometry Pod using the **Zero Pod** button. Remember that the flow head must be left undisturbed on the bench during the zeroing process

2- Click **Start**. Once recording has started ask the volunteer to replace the nose clip and breathe normally through the flow head.

3- Prepare a comment "FVC procedure".

4- Have the volunteer breathe normally for 10 to 20 seconds.

PROCEDURE:

5- Ask the volunteer to **inhale** and then **exhale** as **forcefully**, as **fully** and for **as long** as possible, until no more air can be expired.
6- In the comment box, click Add.
7- Allow the volunteers breathing to return to normal, then click Stop.
8- Your recording should resemble this.
9- Repeat this procedure twice more, so that you have three separate Forced Vital Capacity recordings.

ANALYSIS:

1- Follow the steps below to complete the table.

2- Using the **Waveform Cursor** and the **Marker** tool as necessary, examine each of the three Forced Vital Capacity recordings (FVC).

3- On the Flow channel determine which of the three recordings shows a maximum Peak Inspiratory Flow (PIF).

4- Click to place this data in the **Value panel**. Drag this value into the appropriate cell in the table.

ANALYSIS:

5- Repeat this step to determine the maximum Peak Expiratory Flow (PEF), and enter this into the table also.

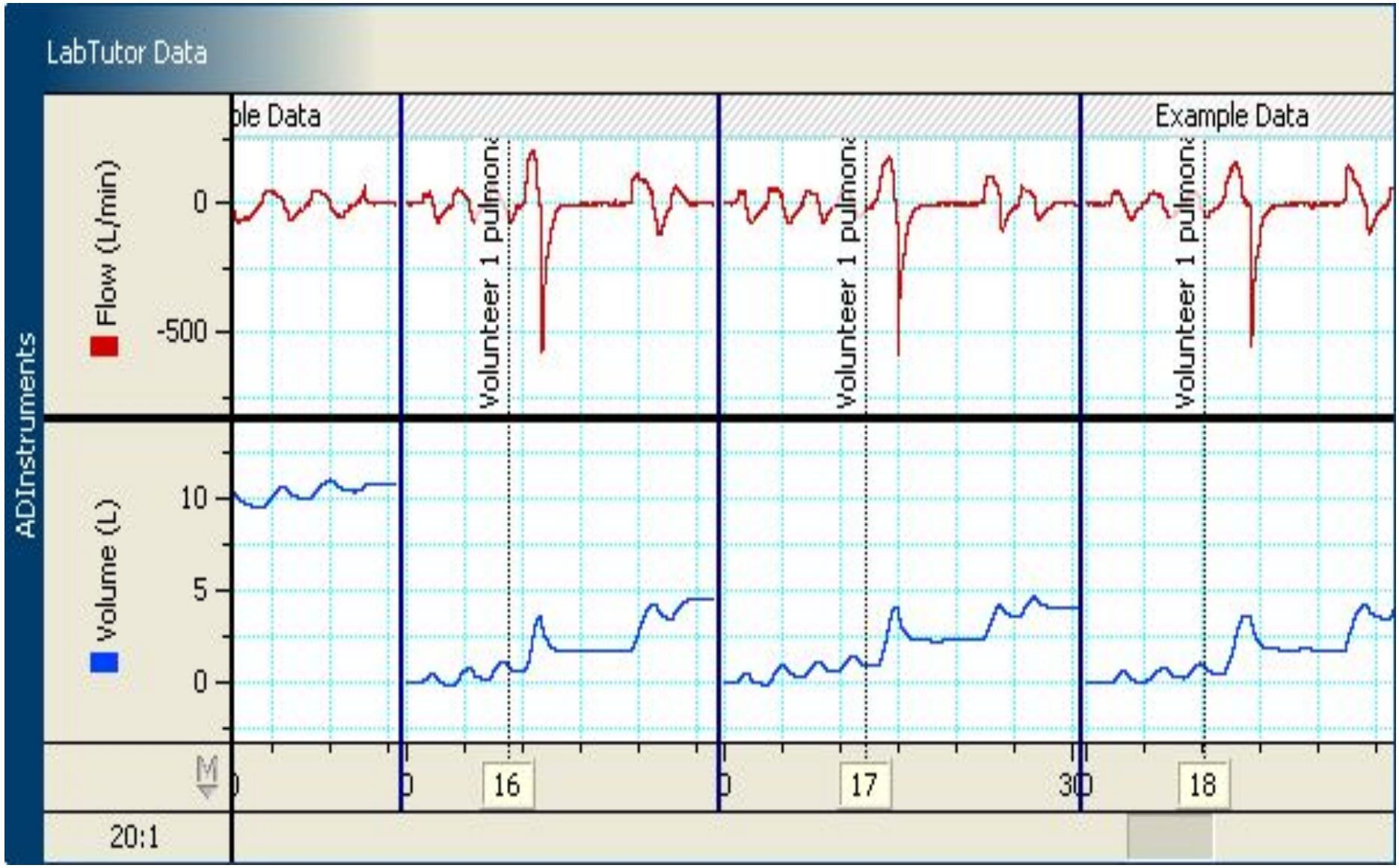
6- On the Volume channel determine which of the three recordings shows a maximal FVC.

7- Place the Marker on the peak inhalation in the Volume channel and move the Waveform Cursor to the maximal expiration also on the Volume channel. Click to place the selected data in the Value panel, and drag the value from the Value panel into the FVC cell of the table.

ANALYSIS:

8- Using the same recording as gave a maximal FVC measure the Forced Expired Volume in 1 second (FEV_1). Place the Marker on the peak inhalation in the Volume channel, move the pointer to a time 1.0 s from the peak. Click to place the selected data in the Value panel, and drag the value from the Value panel into the FVC cell of the table.

9- The ratio of FEV_1 to FVC expressed as a percentage, will be calculated by LabTutor for you.


LabTutor Data
ADInstruments
ble Data
Example Data
Flow (L/min)
0
-500
volunteer 1 pulmoni
volunteer 1 pulmoni
volunteer 1 pulmoni
Volume (L)
10
5
0
16
17
18
20:1

RESULT:

Pulmonary Function Variables				
Respiratory Parameter	Abbreviation	Units	Experimental value	Predicted value
Peak Inspiratory Flow	PIF	L / min		
Peak Expiratory Flow	PEF	L / min		
Forced Vital Capacity	FVC	L		
Forced Expired Volume in 1 second	FEV1	L		
% FVC expired in 1 second	FEV1/FVC x 100			

THANK YOU

How do you define vital capacity and how do you measure it?

It is the total amount of air exhaled after maximal inhalation. The value is about 4800mL and it varies according to age and body size. It is calculated by summing tidal volume, inspiratory reserve volume, and expiratory reserve volume. VC = TV+IRV+ERV.

What are the precautions to be taken during the measurement of vital capacity

- Stop taking your breathing medications for a short period.
- Wear loose, comfortable clothing that doesn't squeeze your chest.
- Refrain from eating a large meal at least two hours before the test.
- Avoid any heavy exercise at least 30 minutes before the test

What are average values of vital capacity in case of normal health young men and women

The total lung capacity in normal healthy male is 5700 mL (4200 mL in female), depending on age, distensibility of the lungs, and the presence or absence of respiratory disease. However, lungs operate at about half capacity, and the lung can never be completely deflated.

Vital Capacity is related to which body parameter and what is the relation

Vital Capacity (VC) is related to several body parameters, most notably height, mass, and age. The relationship is that VC tends to increase with height and mass, and decrease with age. Specifically, VC is the total amount of air a person can exhale after a maximal inhalation, and it is influenced by factors like lung size, which is related to body dimensions like height and mass.

What are the important factors due to which vital capacity may be reduced

The VC averages approximately 50 mL/kg in normal adults. However, VC is not specific and may be decreased because of both inspiratory and expiratory muscle weakness and restrictive lung and chest wall diseases

What is the usefulness of this experiment

Pulmonary function tests (PFTs) are tests that show how well your lungs are working. The tests measure lung volume, capacity, rates of flow, and gas exchange. This information can help to diagnose and decide the treatment of certain lung disorders.

What is dead space/define physiological, anatomical and pathological dead space

Dead space represents the volume of ventilated air that does not participate in gas exchange.

The two types of dead space are anatomical dead space and physiologic dead space.

Anatomical dead space is represented by the volume of air that fills the conducting zone of respiration made up by the nose, trachea, and bronchi. This volume is considered to be 30% of normal tidal volume (500 mL); therefore, the value of anatomic dead space is 150 mL.

Physiologic or total dead space is equal to anatomic plus alveolar dead space which is the volume of air in the respiratory zone that does not take part in gas exchange.

The respiratory zone is comprised of respiratory bronchioles, alveolar duct, alveolar sac, and alveoli.

In a healthy adult, alveolar dead space can be considered negligible. Therefore, physiologic dead space is equivalent to anatomical.

.

what is minute respiratory volume? How it is calculated

Minute respiratory volume (MRV), also known as minute ventilation, is the total volume of gas moved into and out of the lungs in one minute.

It's calculated by multiplying the tidal volume (the amount of air inhaled or exhaled in a single breath) by the respiratory rate (breaths per minute).

Calculation:

Minute respiratory volume (MRV) = Tidal Volume (VT) x Respiratory Rate (RR).